

A Clever Label

Multi-sensory VR data visualization for art, productivity and communication

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ABSTRACT

A *Clever Label* is an interactive immersive documentary pilot experience that introduces a novel data visualization mechanic ("Grapho") for curating and presenting connected graph data as simply as a slide deck. Inside the XR experience the audience is guided by a presenter through volumetric video, voice-overs, haptics and subtitles to explore complex data.

CCS CONCEPTS

• **Human-centered computing** → Visualization; Visualization application domains; Information visualization.

KEYWORDS

VR, documentary, historical, indie

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1 INTRODUCTION

A *Clever Label* is an original pilot from studio, Mod as part of long-term research into interactive storytelling, virtual production, and the "Internet of (Showbiz) Things"[Ledwidge and Burrell, 2015]. 3D spatial and kinetic data user interfaces are common in popular culture but far less common in interactive entertainment let alone in professional digital tools. We explore whether VR can support a useful interaction mechanic in engaging an audience around complex data.

Inspired by journalism projects like *The Panama Papers* [Fitzgibbon, 2021], we explore the insight that today's audiences want to reach conclusions on issues by finding information themselves rather than accepting pre-written, one-way narratives.

2 DESIGN AND IMPLEMENTATION

We begin with the design of a minimalist virtual set in which we can prototype interactive data visualisation ideas. 1) A console that triggers planetarium-style big data constellations. 2) A table-top exhibit of more familiar data visualisations like bar charts and 3)

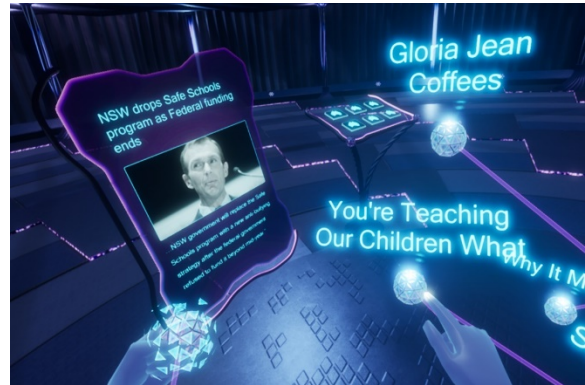


Figure 1: A Clever Label open and closed nodes (Mod)

Grapho – a novel mechanic inspired by using a bullwhip - to quickly flick out and retract virtual graph nodes.

Grapho is the core mechanic in *A Clever Label* and our proposed approach for VR users to grab, manipulate, and navigate complex information nodes – all from curated data served from an API. We provide data "handles" to different queries to allow non-technical presenters to explore subsets of a connected graph with visual, sonic and haptic feedback.

We hypothesise that spatial media allows users time to explore data relationships in ways that traditional media cannot. We aim for an experience as accessible as using a slide deck - contrasting with the complexity of many graph visualization tools.

The key technical processes are 1) managing an arbitrary sized connected graph of JSON and audiovisual media assets like a web browser cache, 2) managing VR user interactions with said data and 3) staging the workflow so that specialist data visualisation tools can be used in parallel to prepare and publish online content. We aim to keep the XR experience navigable for performers in the context of a live real-time show. The project's breakthrough is in providing a presenter-friendly mechanism for accessing 3D graph data visualisations live "on-air" through a pragmatic workflow. Enterprise data tools (e.g. Neo4J Bloom) provide real-time updates to the API used by a TV graphics-friendly VR interface.

3 KEY CHALLENGES

Early peer reviews of the concept on paper and early video documentation elicited feedback such as "it's just a regular VR experience", "the visual quality is not impressive" and "there is no groundbreaking tech". Ahead of completing the pilot it proved difficult to demonstrate how a VR representation of graph data could be any more effective than traditional data representation. COVID19

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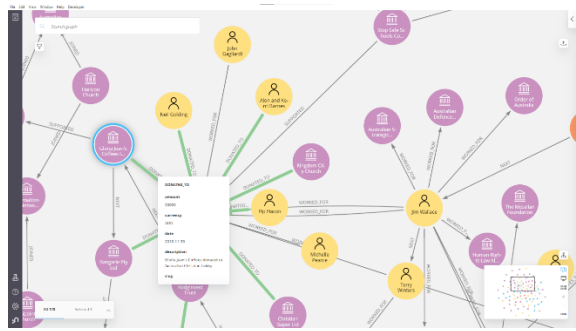


Figure 2: A Clever Label graph data view through Neo4J Bloom (Mod)



Figure 3: A Clever Label Mixed Reality test session

also blocked wider testing plans as part of a Location-Based Experience rollout at festivals based on previous experience deploying touring immersive interactive shows like *ACO Virtual* [Ledwidge, 2014].

In response we pivoted to the release of a free consumer edition for Steam [STEAM, 2022] and itch.io [ITCH.IO, 2022]. This enabled us to garner broader feedback at the cost of further development to support a wider range of VR hardware. We planned to only support Valve Index for our installation - to take advantage of the Index's support for finger tracking and more realistic grab gestures but now support most desktop VR platforms.

Mixed Reality video compositing of the connected graph user experience proved a challenge for how depth is calculated in rendering foreground and background CG elements as the live action presenter moves around. This is an area for further research and improvement in future releases.

4 KEY INSIGHTS

We observed users spending way more time in the experience than expected. We designed the pilot as a 5-15 minute experience but have observed newcomers staying immersed in it for longer than an hour. We observed users occupied with rearranging familiar data points into new structures based on the discovery of new graph linkages. We received positive feedback on the decision to completely omit all user tracking and analytics - aimed at encouraging future use of Grapho for private data exploration.

5 CONCLUSIONS

Grapho's mechanic allows users to move around in VR, following connections between key players and facts, providing a novel way of presenting and exploring information. We plan to extend the product into a generic graph viewing product with capabilities for exploring and bookmarking local data. Grapho is built on an open API that allows data journalists, investigators and production teams to clone our data and populate the experience with their own data sets. We see this as an evolving tool for interactivity within virtual production and virtual world applications.

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